

Mesoscale Gravitational Readability

Persistence, Margins, and Relational Complexes Between Local Curvature and Coarse
Summary

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Abstract

Local curvature evaluation and global coarse-graining answer different gravitational questions. Between them lies a scale-dependent relational layer in which registered source-conditioned responses can form dominance cells, parity faces, seams, low-margin regions, and ablation-complex fingerprints before those structures are compressed by averaging. The revised GCS program now supplies an operational construction for that layer. A full fused scene is solved; each declared source is ablated and re-solved; the comparison geometries are registered to a common support; and the electric-Weyl response

$$\Delta_i E_{ab} = E_{ab}^{(\text{full})} - \tilde{E}_{ab}^{(-i)}$$

is retained as the primary evidence object. The Frobenius amplitude $S_i = \|\Delta_i E\|_{F, \text{ref}}$ is the default direction-agnostic read used to construct the many-source signature network.

This paper defines mesoscale gravitational readability not as a new physical scale or invariant, but as the persistence of declared relational structure across a bounded family of resolutions, thresholds, registrations, and coarse-graining operations. A normalized dominance margin distinguishes robust cell interiors from ambiguous transition neighborhoods. A fixed registered datum skeleton permits before/after comparison under ablation, yielding removed, born, and relabeled strata as an operational source-signature fingerprint. Coarse-graining is treated as a controlled map whose information loss can be measured rather than merely lamented. The central hypothesis is falsifiable: if candidate faces, seams, margins, and complex differences fail to persist under declared perturbations and scale changes, they should not be promoted as mesoscale geometry. If they do persist, they provide a cartographic layer between pointwise tensor data and bulk descriptors without modifying general relativity or introducing physical membranes.

Keywords: general relativity; electric Weyl tensor; tensor ablation; Frobenius norm; mesoscale readability; parity network; dominance margin; relational complex; coarse-graining; source-conditioned signature.

1. Introduction: the middle is now an operational question

Weak-field and initial-data analysis are rich at both ends of description. At the local end, one can evaluate the metric, curvature, electric and magnetic Weyl tensors, matter variables, or geodesic-deviation operators point by point. At the global end, one can average, homogenize, compress, or replace fine structure by effective descriptors suited to the scale of interest [2, 5]. Both operations are lawful. Neither automatically describes how a many-source scene organizes its relational transitions before compression.

The first formulation of mesoscale gravitational readability proposed that retained parity faces, seams, and low-margin regions might occupy this descriptive gap [8]. At that stage the paper relied on assigned weak-field source proxies $Q_i \propto M_i/r_i^3$. The proposal was conceptually useful but operationally incomplete: it assumed source-attributed scalars before establishing how finite-source signatures should be identified in one fused geometry.

The revised GCS program repairs that foundation. GCS I defines registered tensor ablation and the Frobenius parity surface of one declared source [6]. GCS II constructs a many-source network from the registered ablation responses $\Delta_i E_{ab}$ and their default amplitudes $S_i = \|\Delta_i E\|_{F,\text{ref}}$ [7]. The middle layer is therefore no longer merely suggested by proxy pictures. It is a declared family of tensor-derived relational objects.

This paper asks the next question:

Which structures in that family persist across controlled changes of scale, representation, threshold, and context strongly enough to deserve mesoscale descriptive rank?

The word *mesoscale* is used operationally. It does not name a universal length interval. It names a bounded comparison regime between raw pointwise data and a chosen coarse summary in which relational organization remains distinguishable.

2. Scope and object hierarchy

The revised mesoscale program contains five ordered layers:

1. the registered tensor responses $\Delta_i E_{ab}$;
2. the default amplitudes $S_i = \|\Delta_i E\|_{F,\text{ref}}$ and secondary tensor channels;
3. the signature-dominance cells, parity faces, seams, and higher-order nodes of GCS II;
4. the low-margin neighborhoods and fixed-skeleton ablation-complex differences defined here;
5. persistence tests under perturbation, refinement, registration changes, and coarse-graining.

The tensor field remains primary. A surface, cell, seam, or margin is a derived summary of that tensor evidence. A relational complex is a datum skeleton for comparison, not a material partition of space. Mesoscale persistence is a property of a declared analysis pipeline, not an intrinsic label attached to the source or spacetime.

Table 1: Minimal grammar of mesoscale gravitational readability.

Element	Role
Registered scene family	Full and intervention scenes compared on one declared support.
Primary witness	Source-conditioned electric-Weyl response $\Delta_i E_{ab}$.
Default scalar read	Frobenius amplitude $S_i = \ \Delta_i E\ _{F,\text{ref}}$.
Relational structure	Dominance cells, parity faces, seams, nodes, and margin neighborhoods.
Datum skeleton	Fixed registered sampling or cell complex on which labels and strata are compared.
Scale family	Declared resolutions, thresholds, smoothing operators, and registration conventions.
Mesoscale persistence	Survival or controlled deformation of relational structure across that family.

3. Registered signatures and the many-source network

Let $(\Sigma, \gamma_{ab}, K_{ab})$ be a declared full scene with sources $\{S_i\}_{i=1}^N$. For each source i , remove that source, re-solve under the same declared construction, and register the ablated electric-Weyl tensor onto the full-scene support. The source-conditioned response is

$$\Delta_i E_{ab}(x) = E_{ab}^{(\text{full})}(x) - \tilde{E}_{ab}^{(-i)}(x). \quad (1)$$

It is the response of the whole solved scene to the presence of i , including altered nonlinear couplings. It is not a private field component already contained in the full geometry.

Given a declared positive-definite reference metric γ_{ref} , define

$$S_i(x) = \|\Delta_i E(x)\|_{F, \text{ref}} = \left(\gamma_{\text{ref}}^{ac} \gamma_{\text{ref}}^{bd} \Delta_i E_{ab} \Delta_i E_{cd} \right)^{1/2}. \quad (2)$$

The closed signature-dominance cell is

$$\mathcal{C}_i = \{x \in U_{\text{act}} : S_i(x) \geq S_k(x) \text{ for all } k\}, \quad (3)$$

where U_{act} is the declared active support after masks and amplitude-confidence floors. Pairwise retained faces satisfy

$$\overline{\mathcal{F}}_{ij} = \{x \in U_{\text{act}} : S_i = S_j \geq S_k \text{ for all } k\}. \quad (4)$$

Triple and higher-order dominant ties form seam and node strata when the relevant residual gradients are independent. These objects constitute the signature network $\mathcal{N}_{\text{sig}}$ of GCS II.

The mesoscale question begins after this construction. It does not ask whether an equality can be found at one resolution. It asks whether the resulting relational structure survives lawful changes in how the scene is interrogated.

4. Dominance margins and transition neighborhoods

A hard maximum label partitions the active support, but it conceals how decisively the winning signature dominates. Let $S_{(1)}(x) \geq S_{(2)}(x) \geq \dots$ denote the ordered amplitudes at x . Define the normalized dominance margin

$$\mu(x) = \frac{S_{(1)}(x) - S_{(2)}(x)}{S_{(1)}(x) + S_{(2)}(x) + \varepsilon}, \quad (5)$$

where ε is a declared regularization floor tied to numerical or measurement precision. Then $0 \leq \mu \leq 1$ for nonnegative amplitudes. Exact dominant ties satisfy $\mu = 0$, while large margins indicate robust local dominance under the default scalarization.

Definition 1 (Low-margin neighborhood). *For a declared threshold $\tau \in [0, 1]$, the low-margin set is*

$$\mathcal{L}_\tau = \{x \in U_{\text{act}} : \mu(x) \leq \tau\}. \quad (6)$$

The set $\mathcal{L}_\tau$ thickens the zero-thickness parity network into a confidence-aware transition neighborhood. It is not a physical interaction zone. Its width depends on the declared threshold, scalarization, registration, and uncertainty model. Reporting a margin field rather than only a crisp partition prevents numerical near-ties from being mistaken for stable territory.

Secondary tensor channels may reveal ambiguity hidden by amplitude rank. Near repeated eigenvalues, principal directions become unstable even when S_i is smooth. A mesoscale ledger should therefore retain, where relevant, spectral gaps, eigenframe rotation confidence, invariant-polynomial checks, and norm sensitivity alongside μ .

5. Fixed-skeleton relational complexes

The signature network compares all source-conditioned responses in one full scene. Ablation also supports a second mesoscale object: the reorganization of a declared relational complex before and after one intervention.

Let $\mathcal{K}^{(\text{full})}$ and $\mathcal{K}^{(-i)}$ be source-readable complexes evaluated on the same registered datum skeleton D . The skeleton may be a fixed Cartesian lattice, ray family, adaptive mesh with an explicit correspondence, or another declared support. Fixing or registering the skeleton is essential. Without it, a cell that merely moved or was remeshed could be misreported as destroyed and reborn.

For each datum element $d \in D$, let $\ell^{(\text{full})}(d)$ and $\ell^{(-i)}(d)$ record its relational label and stratum type. Define

$$\mathcal{R}_i^- = \{d \in D : \ell^{(\text{full})}(d) \neq \emptyset, \ell^{(-i)}(d) = \emptyset\}, \quad (7)$$

$$\mathcal{R}_i^+ = \{d \in D : \ell^{(\text{full})}(d) = \emptyset, \ell^{(-i)}(d) \neq \emptyset\}, \quad (8)$$

$$\mathcal{R}_i^{\leftrightarrow} = \{d \in D : \ell^{(\text{full})}(d) \neq \ell^{(-i)}(d), \ell^{(\text{full})}(d), \ell^{(-i)}(d) \neq \emptyset\}. \quad (9)$$

The ordered complex difference is

$$\Delta\mathcal{K}_i = (\mathcal{R}_i^-, \mathcal{R}_i^+, \mathcal{R}_i^{\leftrightarrow}). \quad (10)$$

These are removed, born, and relabeled strata under the declared partition rule. Together with the continuous tensor field $\Delta_i E_{ab}$, they form an operational source-signature fingerprint.

The complex difference is not required to be closed, localized, or topologically simple. Its value lies precisely in showing how one intervention reorganizes the relational scaffold globally, including changes far from the ablated source when cross-terms or competitive context are altered.

6. Mesoscale persistence across a scale family

A single extracted network is not yet mesoscale evidence. Let Λ denote a declared family of analysis settings. A parameter $\lambda \in \Lambda$ may encode grid spacing, smoothing width, active-support floor, margin threshold, registration convention, norm choice, or controlled source perturbation. Each setting produces a relational object $\mathfrak{G}_\lambda$, such as a network, low-margin set, or complex difference.

Definition 2 (Mesoscale persistence). *A relational feature $A_{\lambda_0} \subset \mathfrak{G}_{\lambda_0}$ has mesoscale persistence over a declared neighborhood $V \subset \Lambda$ if there exists a correspondence $\Pi_{\lambda_0 \rightarrow \lambda}$ under which the feature survives as a continuously deformed or stably matched structure for $\lambda \in V$, within declared geometric and topological tolerances.*

Persistence is therefore not mere visual resemblance. A reporting protocol should state the feature matcher, distance or overlap measure, topology descriptor, and tolerance. Useful diagnostics include Hausdorff distance for surfaces, Jaccard overlap for thickened regions, label-transition matrices for cells, persistence of seam connectivity, and stability of integrated tensor or margin statistics.

Proposition 1 (Regular deformation of a parity face). *Let $\Psi_{ij}(x, \lambda) = S_i(x, \lambda) - S_j(x, \lambda)$ be continuously differentiable in x and λ . If $\Psi_{ij}(x_0, \lambda_0) = 0$, $\nabla_x \Psi_{ij}(x_0, \lambda_0) \neq 0$, and the dominant-retention inequalities remain strict against all other signatures near (x_0, λ_0) , then the retained pairwise face persists locally as a smooth deformation for sufficiently small changes of λ .*

Proof. The implicit function theorem supplies a smooth local zero-set deformation because the spatial gradient is nonzero. Strict dominance over the remaining signatures is an open condition, so the face remains retained for sufficiently small parameter changes. $\square$

Loss of these conditions is informative. A face can terminate when the active support vanishes, collide with a higher-order tie, lose transversality, fall below confidence, or undergo a genuine topology change. Mesoscale analysis should type these events rather than smoothing them into continuity.

7. Coarse-graining as a controlled information map

Let $\mathcal{A}_\ell$ denote a declared coarse-graining or averaging operator at scale ℓ . It may act on tensor fields, scalar amplitudes, labels, or summary statistics. The relevant question is not whether coarse-graining is good or bad. It is what relational information survives the map.

For a tensor-first lane, compare

$$\mathcal{A}_\ell[\Delta_i E_{ab}] \quad \text{with} \quad \Delta_i \mathcal{A}_\ell[E_{ab}], \quad (11)$$

when both expressions are meaningful under the declared construction. Their difference measures noncommutation between intervention and coarse-graining. Likewise, compare a network extracted after smoothing with the smoothed image of a network extracted before smoothing. Equality should not be assumed because extraction is nonlinear.

Define a schematic relational-loss functional

$$\mathcal{L}_\ell = w_1 d_{\text{geom}} + w_2 d_{\text{top}} + w_3 d_{\text{label}} + w_4 d_{\text{tensor}}, \quad (12)$$

where the distances and weights are declared for the scientific task. The expression is not a universal invariant. It is a reporting grammar for asking which geometry, topology, labels, and tensor information a chosen coarse summary discards.

Two configurations may possess similar total mass, low-order moments, or averaged curvature while differing in $\mathcal{N}_{\text{sig}}$, $\mathcal{L}_\tau$, or $\Delta\mathcal{K}_i$. If those differences persist under controlled normalization, the mesoscale layer carries morphology-sensitive information not represented by the selected bulk descriptor. If they vanish immediately, the coarse descriptor may already be sufficient for that purpose.

8. Static-slice implementation lane and present evidence

The present implementation lane uses time-symmetric, conformally flat static initial data, including exact Brill–Lindquist vacuum scenes and declared extended conformal-density constructions [1, 3]. In the vacuum time-symmetric lane, the electric Weyl tensor is obtained from the spatial curvature under the adopted conventions. Full and ablated scenes are solved on common coordinates, then compared under an explicitly declared registration and reference metric.

Internal benchmarks have established the following limited results:

- registered ablation produces structured global tensor responses rather than simple isolated M/r^3 fields;
- Frobenius-amplitude parity can form compact enclosing GCS shells in tested static scenes;
- many-source FN amplitudes support dominance cells, retained faces, and higher-order tie strata under explicit masks and confidence floors;
- fixed-skeleton before/after comparison yields closed and open delta structures, including relabeled regions that cannot be represented by removed-volume alone;
- scalarization sensitivity can be tested by comparing Frobenius and spectral lanes, while eigenframe channels require degeneracy safeguards.

These are internal benchmark results, not universal claims about dynamical spacetimes. Time evolution, magnetic Weyl structure, moving sources, matter stress, gauge variation, and observational recovery remain open lanes.

9. Failure conditions and falsification discipline

The mesoscale-readability hypothesis weakens when:

- faces, seams, or fingerprints teleport under modest refinement or controlled source perturbation;
- topology is determined primarily by arbitrary masks, thresholds, or registration choices;
- low-margin regions expand without convergence as numerical noise is reduced;
- fixed-skeleton labels fail to stabilize under lawful neighboring partitions;
- morphology discrimination disappears under declared normalization;
- the network adds no information beyond trivial analytic proxy surfaces;
- coarse-grained summaries predict every downstream task as well as the full relational layer;
- no cross-context or cross-scale feature can be matched above appropriate null controls.

Negative results are valuable. A source signature may be globally measurable while its detailed cell morphology drowns rapidly under coarse-graining. That would place a real limit on relational identity rather than invalidate the ablation method. The program should distinguish survival of tensor response, survival of parity geometry, survival of complex topology, and survival of recognizability. These are different evidentiary levels.

10. Relation to dynamics and observation

Mesoscale geometry is not a trajectory theory. A parity face is not a separatrix, a seam is not automatically an attractor, and a low-margin set is not automatically a dynamically neutral zone. Any relation to motion must be tested through geodesic deviation, trajectory integration, stability analysis, or evolution.

Nor is the layer automatically observable. Empirical use requires uncertainty in source rosters, masses, positions, reconstructed geometry, registration, scalarization, masks, and extraction to be propagated into the relational objects. A measurement-aware margin may combine physical near-tie structure with model uncertainty, but measurement noise does not create the root network.

The intended role is cartographic. The relational layer overlays a solved gravitational scene with information about source-conditioned amplitude rank, ambiguity, topology, and response to intervention. It may guide where to inspect dynamics or measurements, but it does not replace them.

11. Evidence register and non-claims

The framework does not modify general relativity, introduce a force, create physical membranes, or establish a preferred decomposition of the full field. It does not claim that dominance cells are sovereign territories, that seams are dynamical boundaries, or that coarse-graining is physically illegitimate. It does not assert universal closure, topology, scale, or observability. The objects are declared diagnostic geometries extracted from registered comparisons.

12. Conclusion

The original mesoscale proposal asked whether a useful descriptive layer exists between pointwise gravitational evaluation and coarse summary. That question remains relevant, but it can now be

Table 2: Claim status of the revised mesoscale program.

Result	Status	Scope
Registered tensor-ablation bundle	Implemented	Exact registration convention must be declared.
FN signature network	Implemented	Static fused scenes on declared active support.
Normalized dominance margin	Defined	Threshold and regularization floor are analysis choices.
Fixed-skeleton complex difference	Implemented diagnostic	Partition and skeleton correspondence must be declared.
Local persistence proposition	Proven sufficient condition	Does not guarantee global topology or branch survival.
Persistent mesoscale geometry	Hypothesis	Requires multi-resolution, perturbation, and registration testing.
Downstream dynamical or observational utility	Open	Must be established separately for each task.
Cross-scale finite-source identity	Seed	No universal invariant yet demonstrated.

posed with sharper machinery.

The revised construction begins with registered electric-Weyl ablation, retains the tensor response $\Delta_i E_{ab}$ as the primary evidence object, and uses the Frobenius amplitude as the default direction-agnostic read. GCS II organizes those amplitudes into source-conditioned dominance cells, parity faces, seams, and nodes. GCS III adds the missing scale discipline: normalized margins, fixed-skeleton ablation-complex fingerprints, persistence across a declared analysis family, and explicit measures of relational loss under coarse-graining.

The middle layer is therefore not defined by a preferred physical length. It is the regime in which relational structure remains distinguishable after local data have been organized but before a chosen summary has erased the distinctions under study. Its existence is empirical and method-dependent. Some features may survive; others may drown.

If the geometry fails to persist, it should be demoted. If it persists, it supplies a reusable cartographic layer for morphology-sensitive comparison, ablation identity tests, scale-aware compression, and targeted downstream analysis. Gravity has not acquired new walls. The description has acquired a sharper middle.

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